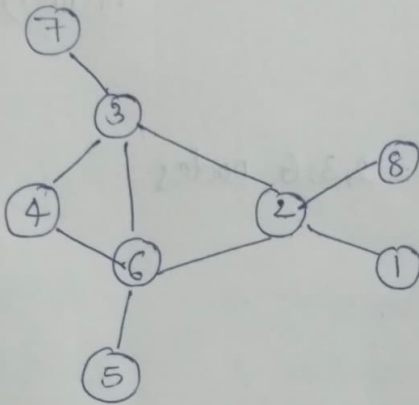


①



a) geodesic distance

b) eccentricity

c) average shortest path

d) radius

e) diameter

f) clustering co-efficient

g) degree centrality

h) Eccentricity centrality

i) closeness centrality

j) betweenness centrality

a)

	1	2	3	4	5	6	7	8	
1	0	1	2	3	3	2	3	2	=16
2	1	0	1	2	2	1	2	1	9
3	2	1	0	1	2	1	1	2	7
4	3	2	1	0	2	2	2	3	8
5	3	2	2	2	0	1	3	3	7
6	2	1	1	1	1	0	2	2	4
7	3	2	1	2	3	2	0	3	3
8	2	1	2	3	3	2	3	0	

b) eccentricity = $\max(d(v_i, v_j))$

$$e(1) = 3$$

$$e(2) = 2$$

$$e(3) = 2$$

$$e(4) = 3$$

$$e(5) = 3$$

$$e(6) = 2$$

$$e(7) = 3$$

$$e(8) = 3$$

$$\begin{aligned}
 \text{c) avg shortest path} &= \frac{\sum_{i=1}^n \sum_{j>i}^n d(v_i, v_j)}{\binom{n}{2}} \\
 &= \frac{84}{28} = 1.92
 \end{aligned}$$

d) radius = 2 (min {ecv})

f) clustering-

$$\text{coefficient (graph)} = 0.208$$

e) diameter = 3 = max {ecv}

$$= \frac{2m}{n(n-1)}$$

g)

1	1
2	4
3	4
4	2
5	1
6	4
7	1
8	1

highest degree

centrality = 2, 3, 6 nodes

h)

1	0.33
2	0.5
3	0.5
4	0.33
5	0.33
6	0.5
7	0.33
8	0.33

i)

CCV) - 1

$$\frac{1}{\sum d(u,v)}$$

1	0.0625
2	0.100
3	0.100
4	0.0714
5	0.0625
6	0.100
7	0.0625
8	0.0625

closeness

centrality = 2, 3, 6 nodes

Eccentric centrality = 2, 3, 6 nodes

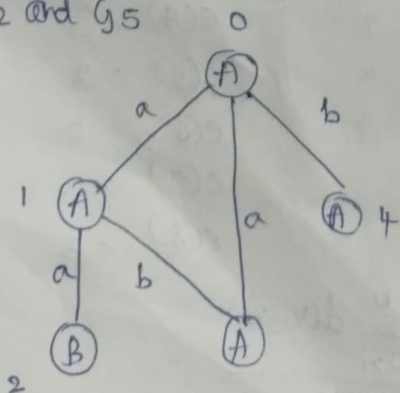
(j) Betweenness centrality for

$$\begin{aligned} 1 &\Rightarrow 0 & 4 &\Rightarrow 0 & 7 &\Rightarrow 0 \\ 2 &\Rightarrow 13.5 & 5 &\Rightarrow 0 & 8 &\Rightarrow 0 \\ 3 &\Rightarrow 8.5 & 6 &\Rightarrow 9.5 \end{aligned}$$

Highest betweenness node = 2

2) G2 and G5

3)



(0, 1, A, a, A)

(1, 2, A, a, B)

(1, 3, A, b, A)

(0, 3, A, a, A)

(0, 4, A, b, A)

Dfs order

1. $A \xrightarrow{a} A$
2. $A \xrightarrow{a} B$
3. Backtrack
4. $A \xrightarrow{b} A$
5. backtrack
6. $A \xrightarrow{b} A$

- | | | |
|---|--------------|---------------|
| 4 | 1. opinion | 7. sentiment |
| | 2. sentiment | 8. sentiment |
| | 3. sentiment | 9. sentiment |
| | 4. opinion | 10. opinion |
| | 5. opinion | 11. opinion |
| | 6. sentiment | 12. sentiment |
| | | 13. sentiment |